Database Design & ER Diagram

Users Table

Stores all user types and roles.

Fields: id (PK), full_name, phone_number, email, role (landlord/renter/investor/manager/vendor/admin), TIN, ID_number, password_hash, created_at, updated_at

Properties Table

Stores property details.

Fields: id (PK), landlord_id (FK→Users), title_deed, address, ward, district, region, GPS_coordinates, property_type, size, rooms, bathrooms, amenities, construction_year, occupancy_permit, photos, virtual_tour_link, created_at, updated_at

Rental Applications Table

Stores tenant rental applications.

Fields: id (PK), property_id (FK→Properties), renter_id (FK→Users), status, lease_start, lease_end, documents, created_at

Lease Agreements Table

Fields: id (PK), application_id (FK→RentalApplications), rent_amount, payment_schedule, signed_at, created_at

Payments Table

Fields: id (PK), payer_id (FK→Users), payee_id (FK→Users), lease_id (FK→LeaseAgreements), request_id (FK→RentAssistance), amount, payment_method, status, created_at

Maintenance Requests Table

Fields: id (PK), property_id (FK→Properties), tenant_id (FK→Users), vendor_id (FK→Users), status, description, created_at, updated_at

Rent Assistance / Microloan Table

Fields: id (PK), renter_id (FK→Users), property_id (FK→Properties), amount_requested, amount_collected, type (gift/loan), status, deadline, repayment_schedule, created_at, updated_at

Contributions Table

Fields: id (PK), request_id (FK→RentAssistance), contributor_id (FK→Users), amount, payment_status, created_at

Messaging / Notifications Table

Fields: id (PK), sender_id (FK→Users), receiver_id (FK→Users), content, type (text/media), status, created_at

Reviews / Ratings Table

Fields: id (PK), reviewer_id (FK→Users), target_id (FK→Users), property_id (FK→Properties), rating, comments, created_at

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Relationships

- Users ↔ Properties (1:N)
- Properties ↔ Rental Applications (1:N)
- Rental Applications ↔ Lease Agreements (1:1)
- Users ↔ Payments (1:N)
- Properties ↔ Maintenance Requests (1:N)
- Rent Assistance ↔ Contributions (1:N)
- Users ↔ Messaging (M:N)
- Reviews ↔ Users/Properties

ER Diagram (Text-based Representation)

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Users (1) ----< Properties
Users (1) ----< RentalApplications >---- Properties (1)
RentalApplications (1) ---- LeaseAgreements (1)
Users (1) ----< Payments >---- Users (1)
Properties (1) ----< MaintenanceRequests >---- Users (Tenant/Vendor)
RentAssistance (1) ----< Contributions >---- Users
Users (M) ----< Messaging >---- Users (M)
Reviews ---- Users/Properties
```

This ER structure can be rendered visually in any ER diagram tool (draw.io, Lucidchart, dbdiagram.io) to create a proper PDF diagram for developers. It represents **all modules, relationships, and key compliance fields** for Panga na Kupangisha.